

Experiment No.7

Aim: To study and implement Identity and Access Management (IAM) practices on AWS/Azure cloud.

Objectives:

- To develop a theoretical understanding of Identity and Access Management (IAM) principles.
- To apply practical knowledge to configure IAM policies, users, and roles on.

Outcomes:

- Demonstrate proficiency in configuring IAM policies, users, groups, and roles on AWS.
- Applied knowledge in implementing IAM practices to enhance security, streamline access management, and adhere to the principle of least privilege.

Software Requirements: Windows/Ubuntu OS, AWS Account.

Theory:

IAM is a crucial aspect of cloud security, allowing users to control access to cloud resources and services. On AWS, IAM involves the creation of policies that define permissions, users who inherit those permissions, and groups and roles to organize users logically. Azure follows a similar structure with role-based access control (RBAC) to assign permissions, users, and groups.

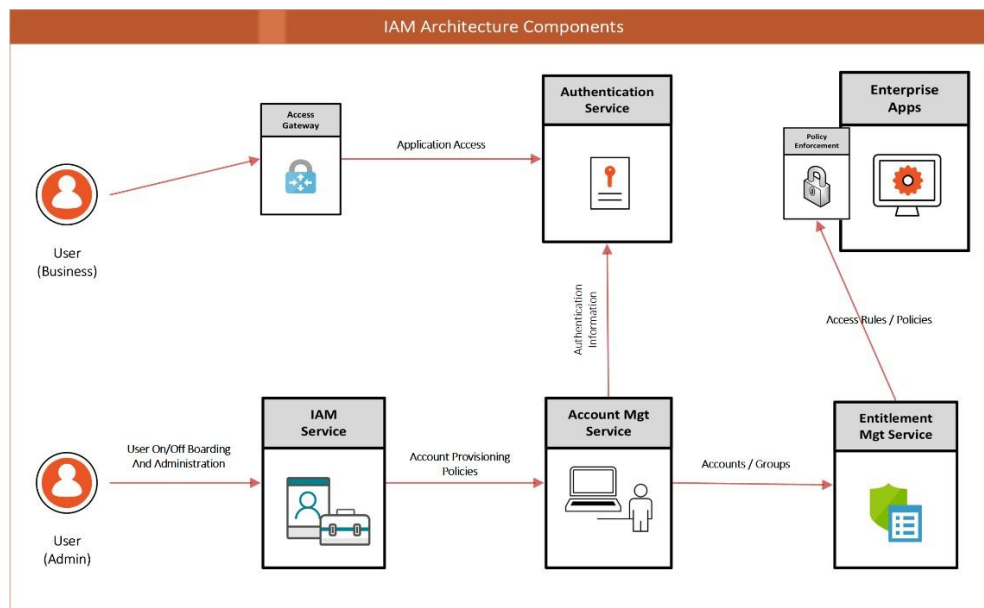


Figure 8 IAM Architecture Components

Key Components:

1. **Users:** Individuals who need access to resources, identified by unique usernames or email addresses.
2. **Groups:** Collections of users with similar access needs, simplifying permission management.
3. **Policies:** Define sets of permissions that can be assigned to users, groups, or services, decoupling permissions from identities.

4. **Authentication:** Process of verifying a user's identity, typically using credentials like passwords or multi-factor authentication.
5. **Authorization:** Process of determining what actions a user is allowed to perform, based on their assigned permissions.

How It Works:

1. **User Authentication:** A user attempts to access a resource and provides their credentials.
2. **Identity Verification:** IAM validates the user's identity against its database.
3. **Policy Evaluation:** IAM checks the user's assigned policies to determine if the requested action is allowed.
4. **Access Grant or Deny:** If the policy permits the action, IAM grants access to the resource. Otherwise, access is denied.

Benefits of IAM:

- **Centralized Control:** Manage user identities and permissions from a single location.
- **Enhanced Security:** Enforce granular access control to protect sensitive resources.
- **Reduced Complexity:** Simplify permission management with roles and groups.
- **Improved Compliance:** Audit user access and enforce security best practices.
- **Scalability:** Adapt to changing security requirements as your organization grows.

Stepwise Procedure to Implement PaaS using AWS Elastic Beanstalk:

Step 1: Sign into AWS Console

- Log in to the AWS Management Console using your AWS account credentials.

Step 2: Access IAM Service

- Navigate to the IAM service in the AWS Console.

Step 3: Create IAM Users

- Create IAM users with associated access keys for programmatic access.
- Assign permissions to users using IAM policies.

Step 4: Create IAM Groups

- Organize users into logical groups based on their roles or responsibilities.
- Assign policies to groups to grant permissions collectively.

Step 5: Define IAM Roles

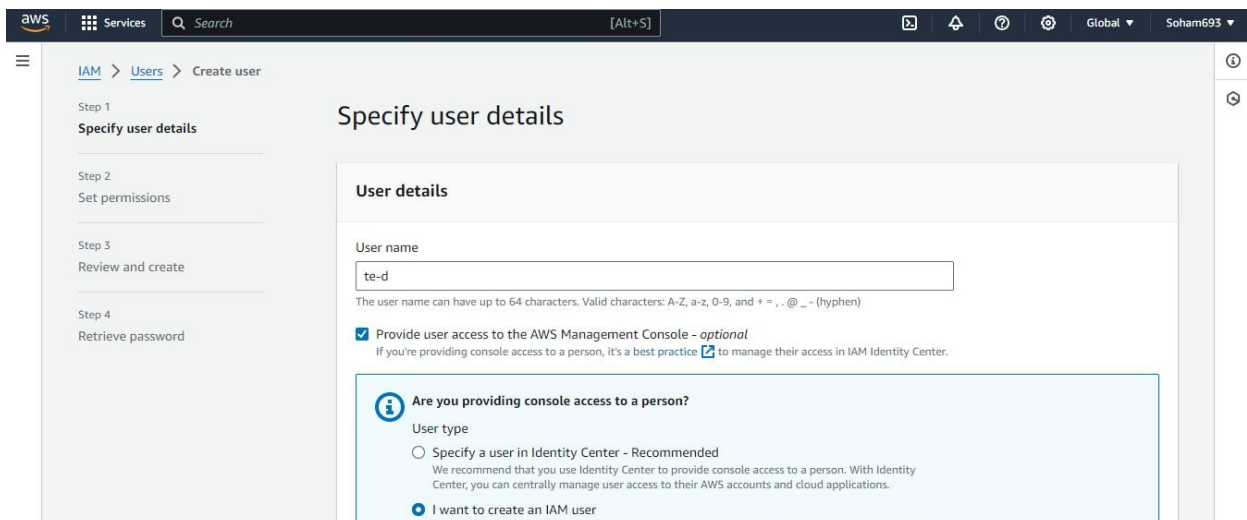
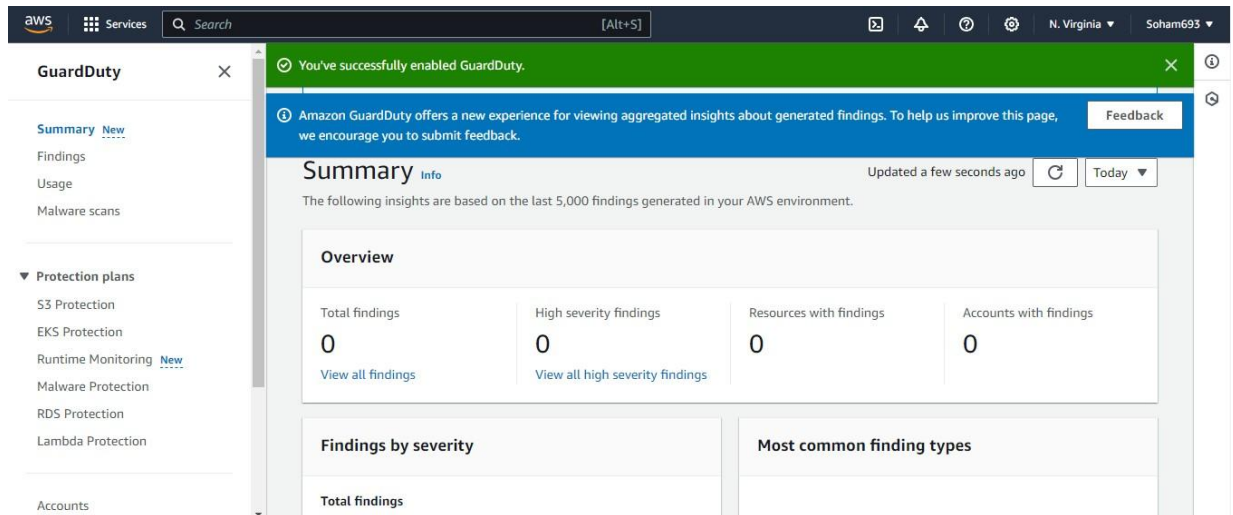
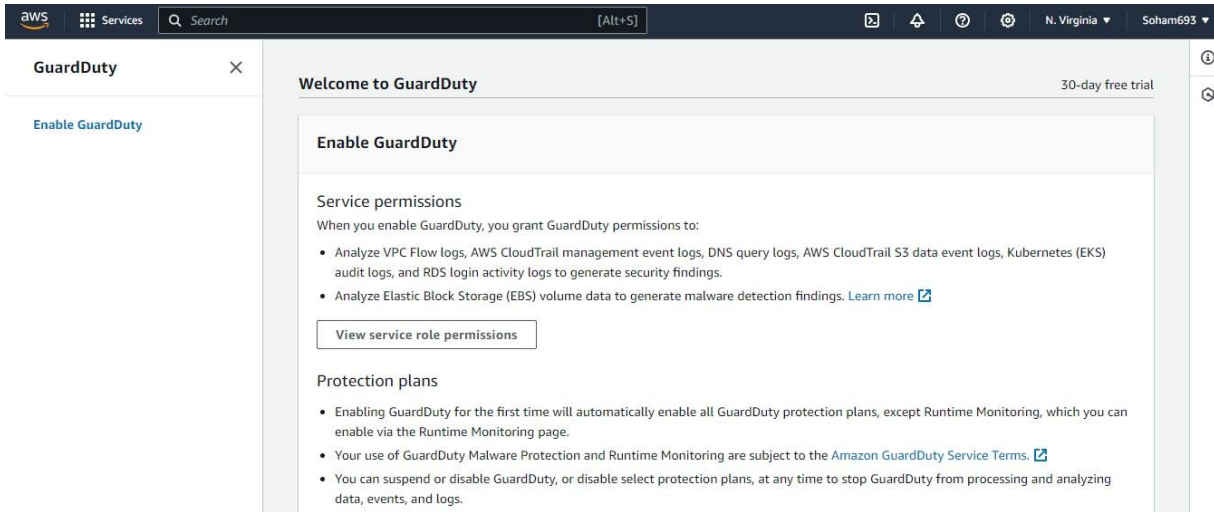
- Create IAM roles with specific permissions.
- Define trust relationships to establish who can assume the role.

Step 6: Attach Policies to Users and Groups

- Attach IAM policies to users or groups to grant necessary permissions.
- Review and modify policies based on security requirements.

Step 7: Test IAM Policies

- Test IAM policies by attempting to access AWS resources using the configured users and roles.
- Troubleshoot and refine policies as needed.



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Permissions policies (1/1183)

Choose one or more policies to attach to your new user.

s3

Filter by Type

All types

12 matches

Policy name	Type	Attached entities
☐ AmazonDMSRedshiftS3Role	AWS managed	0
☒ AmazonS3FullAccess	AWS managed	1
☐ AmazonS3ObjectLambdaE...	AWS managed	0
☐ AmazonS3OutpostsFullAcc...	AWS managed	0
☐ AmazonS3OutpostsReadO...	AWS managed	0
☐ AmazonS3ReadOnlyAccess	AWS managed	0
☐ AWSBackupServiceRolePol...	AWS managed	0
☐ AWSBackupServiceRolePol...	AWS managed	0

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IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1183)

Choose one or more policies to attach to your new user.

Filter by Type

aws

Services

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search for services, features, blogs, docs, and more

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Identity and Access Management (IAM)

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Access reports

Access analyzer

Archive rules

Analyzers

Settings

Credential report

Users > senior_engineer

Summary

User ARN

arn:aws:iam::066488174122:user/senior_engineer

Path

/

Creation time

2022-04-10 22:36 UTC+0530

Permissions

Groups

Tags

Security credentials

Access Advisor

Permissions policies (1 policy applied)

Add permissions

Policy name	Policy type
Attached directly	
AmazonS3FullAccess	AWS managed policy

Permissions boundary (not set)

Manage MFA device

Choose the type of MFA device to assign:

☒ Virtual MFA device

Authenticator app installed on your mobile device or computer

☐ U2F security key

YubiKey or any other compliant U2F device

☐ Other hardware MFA device

Gemalto token

For more information about supported MFA devices, see [AWS Multi-Factor Authentication](#)

Cancel

Continue

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Identity and Access Management (IAM)

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Set up virtual MFA device

1. Install a compatible app on your mobile device or computer

See a [list of compatible applications](#)

2. Use your virtual MFA app and your device's camera to scan the QR code



Cancel

Previous

Assign MFA

Set up virtual MFA device

✓

You have successfully assigned virtual MFA

This virtual MFA will be required during sign-in.

Close

Introducing the new Users list experience

We've redesigned the Users list experience to make it easier to use. [Let us know what you think.](#)

IAM > Users

Users (2) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.



Delete

Add users

Find users by username or access key



1



☐	User name	Groups	Last activity	MFA	Password a...	Active
☐	junior_engineer	None	Never	None	✓ 16 minutes ago	-
☐	senior_engineer	None	Never	Virtual	✓ 13 minutes ago	-

Amazon S3

Buckets

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

Access analyzer for S3

Block Public Access settings for this account

Storage Lens

Dachhnaarde

Read the S3 resources page for documentation and technical content.

[Amazon S3](#) > [Buckets](#) > pqrst

pqrst [Info](#)

Objects

Properties

Permissions

Metrics

Management

Access Points

Permissions overview

Access

Bucket and objects not public

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#) 

Edit

Delete



Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. [Learn more about using Amazon S3 Block Public Access](#) 

Copy the policy

```
{
  "Version": "2012-10-17",
  "Id": "123",
  "Statement": [
    {
      "Sid": "",
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": "arn:aws:s3:::pqrst/confidential/*",
      "Condition": {
        "Null": { "aws:MultiFactorAuthAge": "true" }
      }
    },
    {
      "Sid": "",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": "arn:aws:s3:::pqrst/*"
    }
  ]
}
```

Save Changes

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/

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Permissions

Groups

Tags

Security credentials

Access Advisor

Sign-in credentials

Summary

• Console sign-in link: <https://066488174122.signin.aws.amazon.com/console>

• MFA is required when signing in. [Learn more](#)

Console password

Enabled (last signed in Today) | [Manage](#)

Assigned MFA device

arn:aws:iam::066488174122:mfa/senior_engineer (Virtual) | [Manage](#)

Signing certificates

None

Upload succeeded
View details below.

Destination

s3://pqrst/confidential/

Succeeded

1 file, 18.1 KB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 Total, 18.1 KB)

Find by name

< 1

Name	Folder	Type	Size	Status	Error
LES_SEM-8_CCL.docx	-	application/vnd.openxmlformats-officedocument.wordprocessingml.document	18.1 KB	Succeeded	-

Conclusion:

Thus, we are able to acquire a solid understanding of Identity and Access Management (IAM) principles and practices on AWS. By configuring IAM users, groups, and roles, we gained proficiency in securing AWS resources and managing access effectively.